ES32079 Financial Derivatives
Problem Set 1 Solutions

Question 1

Give an example of a **hedger**, a **speculator**, and an **arbitrageur** in derivatives markets.

Why might a hedger sometimes be accused of "speculating"? What controls should firms have in place?

Solution

- **Hedger:** A wheat farmer expects to sell wheat in three months and is concerned that its price may fall. The farmer shorts wheat futures.

  If wheat prices fall, the gain on the futures position offsets some or all of the lower cash-market revenue. The hedge stabilises the selling price, although basis risk may remain.

- **Speculator:** A trader expects crude-oil prices to rise and buys oil futures without having an existing oil exposure.

  The trader profits if the futures price rises and loses if it falls. The objective is to take risk in anticipation of a return.

- **Arbitrageur:** Suppose gold is priced at \$2,000 per ounce and carrying costs imply a fair one-year futures price of \$2,120, but the market futures price is \$2,150.

  The arbitrageur can buy gold, finance and carry it for one year, and short the futures contract. At maturity, the gold is delivered for \$2,150, generating an arbitrage profit of

  $$\$2{,}150 - \$2{,}120 = \$30$$

  per ounce, before transaction costs.

A hedge may become speculative if the derivative position does not match the underlying exposure. This can occur when the firm:

- hedges more than its actual exposure;

- trades without an underlying exposure;
- changes the hedge according to market forecasts; or
- uses instruments that introduce additional risks.

Firms should therefore have:

- a written hedging policy;
- documentation linking each derivative to an underlying exposure;
- position limits and appropriate authorisation;
- independent risk monitoring and valuation; and
- regular reporting and reconciliation.

A hedge should be evaluated by how effectively it reduces the risk of the **combined position**, not by whether the derivative makes a profit.

Question 2

Which of the following instruments are OTC vs exchange-traded?

- Futures on crude oil
- A customized forward contract on jet fuel
- An S&P 500 index option
- A 5-year interest rate swap

What are the advantages and disadvantages of OTC vs exchange? Why do most FX and interest rate derivatives trade OTC instead of on exchanges?

Solution

Instrument	**Typical market**
Crude-oil futures	Exchange-traded
Customised jet-fuel forward	OTC
Standard S&P 500 index option	Exchange-traded
Five-year interest-rate swap	OTC

Exchange-traded derivatives have standardised contract sizes, maturities, and terms. Their main advantages are:

- transparent market prices;

- generally high liquidity;
- central clearing and daily margining; and
- reduced and actively managed counterparty risk.

Their main disadvantage is limited flexibility: the available contracts may not match the amount, maturity, or underlying exposure that a firm needs to hedge.

OTC derivatives are negotiated between counterparties. Their main advantage is customisation: the parties can choose the notional amount, maturity, underlying asset, and payment terms.

Their disadvantages may include:

- lower liquidity for customised contracts;
- less public price transparency;
- more complex documentation and valuation; and
- counterparty risk, particularly for uncleared contracts.

Many FX and interest-rate derivatives trade OTC because firms and financial institutions often need contracts that match precise currencies, amounts, dates, interest-rate conventions, and cash flows. OTC dealer markets can provide this flexibility and can also be highly liquid for commonly traded products.

Some standardised OTC derivatives are centrally cleared, while uncleared contracts may be subject to collateral, margin, and reporting requirements.

Question 3

A farmer expects to harvest 50,000 bushels of wheat in 3 months. Wheat futures (contract size 5,000 bushels) for delivery in 3 months trade at \$5.20 per bushel.

1. What hedge should the farmer take today?
2. How many contracts are required?
3. If the spot price turns out to be \$4.80 at harvest and the futures settle at \$4.85, what is the effective price the farmer receives?

Why is the hedge not perfect?

Solution

Because the farmer will **sell** wheat in three months and is concerned that its price may fall, the farmer should take a **short futures hedge**.

The required number of contracts is

$$N = \frac{\text{quantity hedged}}{\text{contract size}} = \frac{50{,}000}{5{,}000} = \boxed{10 \text{ short contracts}}.$$

At harvest, the farmer sells the physical wheat in the spot market:

$$\text{Spot revenue} = 50{,}000(\$4.80) = \$240{,}000.$$

The short futures position gains

$$\$5.20 - \$4.85 = \$0.35$$

per bushel. Therefore,

$$\text{Futures gain} = 50{,}000(\$0.35) = \$17{,}500.$$

The total proceeds are

$$\$240{,}000 + \$17{,}500 = \$257{,}500.$$

Hence, the effective price per bushel is

$$\text{Effective price} = \frac{\$257{,}500}{50{,}000} = \boxed{\$5.15 \text{ per bushel}}.$$

Equivalently,

$$\text{Effective price} = S_T + (F_0 - F_T) = 4.80 + (5.20 - 4.85) = \$5.15.$$

The hedge is imperfect because of **basis risk**. The closing basis is

$$b_T = S_T - F_T = 4.80 - 4.85 = -\$0.05.$$

Thus,

$$\text{Effective price} = F_0 + b_T = 5.20 - 0.05 = \$5.15.$$

The effective price differs from the initial futures price because the spot and futures prices do not move perfectly together. Differences in quality, location, timing, transaction costs, and the quantity actually harvested may also reduce hedge effectiveness.

Question 4

Define the **basis** and explain why it matters in a hedge.

Spot price of corn today = \$3.00, futures price (3 months) = \$3.10. In 3 months, spot = \$2.70, futures = \$2.75.

- What was the basis at hedge initiation and at hedge close?
- How did the basis risk affect the hedge outcome?

Why does the basis tend to converge to zero at maturity?

Solution

The basis is defined as

$$b_t = S_t - F_t,$$

where S_t is the spot price and F_t is the futures price.

The initial basis is

$$b_0 = S_0 - F_0 = 3.00 - 3.10 = \boxed{-\$0.10}.$$

The closing basis is

$$b_T = S_T - F_T = 2.70 - 2.75 = \boxed{-\$0.05}.$$

The basis has **strengthened**, because it increased from $-\$0.10$ to $-\$0.05$.

For a short hedger, such as a corn farmer, the effective selling price is

$$\text{Effective price} = S_T + (F_0 - F_T).$$

Therefore,

$$\text{Effective price} = 2.70 + (3.10 - 2.75) = \boxed{\$3.05}.$$

Equivalently,

$$\text{Effective price} = F_0 + b_T = 3.10 - 0.05 = \$3.05.$$

If the basis had remained at $-\$0.10$, the effective price would have been

$$3.10 - 0.10 = \$3.00.$$

Thus, the strengthening basis improves the short hedger's effective selling price by \$0.05 per bushel. For a long hedger, the same basis strengthening would increase the effective purchase price and would therefore be unfavourable.

The uncertainty about the closing basis is called **basis risk**. It means that a futures hedge does not normally guarantee the exact initial futures price.

At the delivery date, the futures price and spot price for the same deliverable asset, location, and delivery terms must converge:

$$F_T \to S_T \qquad \Rightarrow \qquad b_T = S_T - F_T \to 0.$$

Otherwise, traders could buy in the cheaper market and sell in the more expensive market, creating an arbitrage opportunity.

Question 5

An airline wants to hedge jet fuel costs, but there are no jet fuel futures. It considers using crude oil futures.

What are the risks in this strategy? What factors determine whether cross-hedging is effective?

Solution

Because the airline will **buy** fuel in the future and is concerned that prices may rise, it should take a **long position** in crude-oil futures.

This is a **cross-hedge** because the asset being hedged, jet fuel, is different from the asset underlying the futures contract, crude oil.

The main risks are:

- **Imperfect correlation:** jet-fuel and crude-oil prices may not move together.
- **Refining-spread risk:** changes in refining costs and margins can cause jet-fuel prices to move differently from crude-oil prices.
- **Timing mismatch:** the futures maturity may not coincide with the airline's fuel-purchase date.
- **Location and benchmark mismatch:** the crude-oil contract may refer to a different region or pricing benchmark.
- **Margin risk:** daily futures losses may create cash-flow requirements even if the physical hedge is economically effective.

The cross-hedge is more effective when changes in jet-fuel prices are strongly and consistently related to changes in crude-oil futures prices.

A high and stable correlation, appropriate contract maturity, similar pricing locations, and good futures-market liquidity generally make the cross-hedge more effective. However, because the two commodities are not identical, some basis risk remains.

Question 6

Historical regression of ΔS on ΔF gives slope $h^* = 0.8$. If the airline's fuel exposure is 1,000,000 gallons and each futures contract = 42,000 gallons, how many contracts should it use?

Why might h^* be less than 1? Can over-hedging ever make sense?

Solution

The optimal number of futures contracts is

$$N^* = h^* \frac{Q_A}{Q_F},$$

where Q_A is the size of the exposure and Q_F is the quantity covered by one futures contract.

Therefore,

$$N^* = 0.8 \left(\frac{1{,}000{,}000}{42{,}000} \right) = 19.05.$$

Because futures contracts must be traded in whole numbers, the airline should use approximately

$$\boxed{\text{19 long crude-oil futures contracts}}.$$

The position is long because the airline will buy fuel in the future and is concerned that fuel prices may rise.

The minimum-variance hedge ratio is

$$h^* = \rho \frac{\sigma_S}{\sigma_F}.$$

In the present case, because $h^* = 0.8$, using substantially more than 19 contracts would move the airline away from the estimated minimum-variance hedge.

A hedge ratio greater than one can nevertheless be appropriate in a different case. For example, if the spot exposure is more volatile than the futures price and correlation is sufficiently high, then

$$\rho \frac{\sigma_S}{\sigma_F} > 1.$$

In that case, a futures position larger than the physical quantity may minimise the variance of the hedged position. It is therefore not necessarily speculative.

However, taking a position larger than the estimated minimum-variance hedge because of a market forecast introduces additional risk and should be regarded as speculation rather than pure hedging.